E-07-quantitative-compressive v1: quantitative estimate for D_t eta_- with spectral frame dynamics

ZFL Lab

September 3, 2026

MISSION: Convert $D_t\eta_-$ into quantitative verifiable estimate, attacking W_- bottleneck. INPUT: E-05 spectral: $\omega^T S\omega = \sum \lambda_i \omega_i^2$, $\sum \lambda_i = 0$. E-06: $\eta_- = \sum_{\lambda_i<0} \omega_i^2/|\omega|^2$, $D_t\xi = P_{\xi^\perp} S\xi + \Omega\xi + \nu\Delta\xi + 2\nu\nabla \log|\omega| \cdot \nabla\xi$. $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ frozen.

FORBIDDEN: η_- large $\not\Rightarrow W_- \in L_t^1$. $D_t\eta_- < 0 \not\Rightarrow \eta_-$ never regrows. $C_E^{weak} \not\Rightarrow C_E^{strong}$. Omitting $D_t e_i$ term. Claiming quantitative control without e_i dynamics.

AUDIT:

Setup: Where $|\omega| > 0$, $Se_i = \lambda_i e_i$, $\lambda_1 \geq \lambda_2 \geq \lambda_3$, $\sum \lambda_i = 0$. Let $I_-(x,t) = \{i : \lambda_i < 0\}$. Assume for estimate case (ii) $\lambda_1 > 0, \lambda_2 < 0, \lambda_3 < 0$ (2 compressive) typical near blow-up candidate; case (i) similar with $I_- = \{3\}$.

Define $\eta_- = \sum_{i\in I_-}(\xi \cdot e_i)^2 = \sum_{i\in I_-} \eta_i$, $\eta_i = (\xi \cdot e_i)^2$.

Exact $D_t\eta_-$ contains frame derivative:

$\eta_i = (\xi \cdot e_i)^2$, so

$$D_t\eta_i = 2(\xi \cdot e_i)D_t(\xi \cdot e_i) = 2(\xi \cdot e_i)[(D_t\xi) \cdot e_i + \xi \cdot D_t e_i].$$

Thus

$$\boxed{D_t\eta_- = 2\sum_{i\in I_-}(\xi \cdot e_i)[(D_t\xi) \cdot e_i + \xi \cdot D_t e_i] + \sum_{i\in I_-}(\xi \cdot e_i)^2 D_t \mathbf{1}_{I_-}}.$$

Last term: index set $I_-(x,t)$ changes when λ_i crosses 0, discontinuous. For quantitative estimate we work on region where sign pattern fixed, or include measure of crossing as OPEN.

Insert $D_t\xi$ complete form:

$$D_t\xi = S\xi - (\xi^T S\xi)\xi + \Omega\xi + \nu\Delta\xi + 2\nu(\nabla \log|\omega| \cdot \nabla)\xi.$$

Let $\alpha = \xi^T S\xi = \sum \lambda_i \eta_i$.

Then $(D_t\xi) \cdot e_i = (S\xi) \cdot e_i - \alpha(\xi \cdot e_i) + (\Omega\xi) \cdot e_i + \nu(\Delta\xi) \cdot e_i + 2\nu(\nabla \log|\omega| \cdot \nabla\xi) \cdot e_i$.

Since $(S\xi) \cdot e_i = \lambda_i(\xi \cdot e_i)$ (using S symmetric), we get

$$(S\xi) \cdot e_i - \alpha(\xi \cdot e_i) = (\lambda_i - \alpha)(\xi \cdot e_i).$$

Thus part without rotation/diffusion/frame:

$$D_t\eta_-^S = 2\sum_{i\in I_-}(\lambda_i - \alpha)\eta_i.$$

This is exact and key: if α is average $\sum \lambda_j \eta_j$, then $\lambda_i - \alpha$ negative when η_- large and λ_i most negative.

Estimate: Let $\lambda_{min} = \lambda_3$, $\lambda_{max} = \lambda_1$. Then $\alpha = \lambda_1(1-\eta_-) + \bar{\lambda}_-\eta_-$ with $\bar{\lambda}_- < 0$ average compressive. If $\eta_- \to 1$, $\alpha \to \bar{\lambda}_- < 0$, then $\lambda_i - \alpha \to \lambda_i - \bar{\lambda}_-$ may be small. If $\eta_- \to 0$, $\alpha \to \lambda_1 > 0$, then $\lambda_i - \alpha \approx \lambda_i - \lambda_1 \le 0$ strongly negative for $i \in I_-$. So S term drives $D_t\eta_-^S < 0$ when η_- small? Check sign:

If η_- small, $\alpha \approx \lambda_1 > 0$, then for $i \in I_-$, $\lambda_i - \alpha < 0$, so $D_t\eta_-^S = 2\sum_{i\in I_-}(\lambda_i - \alpha)\eta_i < 0$, i.e., compressive fraction decreases further. So S alone tends to push $\eta_- \to 0$. Good.

If $\eta_- \approx 1$, $\alpha \approx \bar{\lambda}_- < 0$, then for $i \in I_-$ near average, $\lambda_i - \alpha \approx 0$, small drift. So near compressive state, S term neutral.

Now full quantitative inequality attempt:

$$\boxed{D_t\eta_- = 2\sum_{i\in I_-}(\lambda_i - \alpha)\eta_i + 2\sum_{i\in I_-}(\xi\cdot e_i)(\Omega\xi\cdot e_i) + 2\nu\sum_{i\in I_-}(\xi\cdot e_i)(\Delta\xi\cdot e_i) + 4\nu\sum_{i\in I_-}(\xi\cdot e_i)(\nabla\log|\omega|\cdot\nabla\xi\cdot e_i) + 2\sum_{i\in I_-}}$$

We bound rotation term: $|(\xi\cdot e_i)(\Omega\xi\cdot e_i)| \le |\Omega|$ since Ω skew, $|\Omega| \sim |\omega|/2$? Actually $|\Omega| \le |\nabla u|$. So term $\le C|\nabla u|\sqrt{\eta_-}$.

Crucial uncontrolled term: $\xi\cdot D_t e_i$. e_i dynamics depends on D_tS: D_te_i involves $(\lambda_i - \lambda_j)^{-1}e_j^T(D_tS)e_i$ (standard perturbation). D_tS contains $\nabla D_t u$ = pressure Hessian + nonlinear. No control from $\mathcal{C}_{PROVEN}$. So term OPEN/BLOCKED.

Diffusion terms: $\nu(\Delta\xi)$ after integration against $|\omega|^2$ yields $-\nu|\nabla\xi|^2$ dissipative for η_-? Need weighted estimate.

Consider weighted functional for quantitative control:

$$\Phi(t) = \int \Phi(\eta_-)|\omega|^2 dx,\ \Phi \text{ increasing, e.g. } \Phi(s) = s.$$

Then

$$\frac{d}{dt}\int \eta_-|\omega|^2 = \int (D_t\eta_-)|\omega|^2 + \eta_- D_t|\omega|^2 + \text{transport}.$$

Using $\frac{1}{2}D_t|\omega|^2 = \alpha|\omega|^2 + \frac{\nu}{2}\Delta|\omega|^2 - \nu|\nabla\omega|^2$, we get coupling.

Formal target inequality:

$$\boxed{\frac{d}{dt}\int \eta_-|\omega|^2 \le \int [A\eta_- + B(1-\eta_-)]|\omega|^2 + \text{diffusion} + \int |\omega|^2\xi\cdot D_te_i \text{ terms}}.$$

Where $A = 2(\bar{\lambda}_- - \alpha)$ etc.

No closure without D_te_i bound. Hence:

RESULT classification:

We obtain exact identity for $D_t\eta_-$ including D_te_i and crossing of I_-.

We obtain sign structure: $D_t\eta_-^S \le 0$ when η_- small (S pushes to stretching), but near $\eta_- \approx 1$, drift small.

We do NOT obtain $W_- \in L_t^1$.

Quantitative obstruction:

$$\boxed{D_te_i \text{ requires } D_tS = \frac{1}{2}(D_t\nabla u + D_t\nabla u^T) \text{ which contains pressure Hessian } \nabla^2 p \text{ and is not controlled by } \mathcal{C}_{PRO}}$$

Thus $D_t\eta_-$ estimate has OPEN term.

OUTPUT:

$$\boxed{D_t\eta_- = 2\sum_{i\in I_-}(\lambda_i - \alpha)\eta_i + \text{rotation} + \text{diffusion} + 2\sum_{i\in I_-}(\xi\cdot e_i)(\xi\cdot D_te_i) \text{ EXACT}}$$

$\boxed{D_t\eta_-^S \leq 0 \text{ when } \eta_- \text{ small, i.e., } S \text{ tends to eliminate compressive fraction}}$

$\boxed{\text{Persistent } \eta_- \to 1 \text{ requires balancing } S\text{-drift plus } D_t e_i + \Omega + \nu \text{ diffusion}}$

$\boxed{\text{No quantitative closure: } D_t e_i = \text{OPEN/BLOCKED, needs } D_t S}$

$\boxed{\text{Potential bridge: If } D_t e_i \text{ controlled, then } \int \eta_- |\omega|^2 \text{ decreasing on average, tension with } C_E^{weak}}$

Classification: PARTIAL/OPEN — identity useful, but spectral frame term without control.
$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ untouched. No C_E^{strong} claimed.
Next: audit $D_t S$ structure (pressure Hessian) as E-08 if want to close frame term, or accept PARTIAL and freeze E-07.